

A. The behaviour handed over

Input index ι	a	b	c_{in}	sum	carry
0	0	0	0	0	0
1	1	0	0	1	0
2	0	1	0	1	0
3	1	1	0	0	1
4	0	0	1	1	0
5	1	0	1	0	1
6	0	1	1	0	1
7	1	1	1	1	1

The stated relation $a + b + c_{in}$, enumerated over its eight states. This table is the whole of the input: no circuit is supplied and no gate is named.

B. The same behaviour as an index set

	0	1	2	3	4	5	6	7
sum	0	1	1	0	1	0	0	1
carry	0	0	0	1	0	1	1	1

Index $\iota = a + 2b + 4c_{in}$

Only the positions of the ones are kept: sum $\mathcal{I} = \{1, 2, 4, 7\}$, carry $\mathcal{I} = \{3, 5, 6, 7\}$
The carry's set is covered exactly by 11^* , 1^*1 , *11 , with anchors 3, 5, 6.
That is the three-input majority index set of Section S3.1, reached here from arithmetic rather than from voting.

C. Which coordinates the behaviour responds to

carry	0	1	2	3	4	5	6	7
	0	0	0	1	0	1	1	1

A single-bit flip that crosses the set makes that coordinate essential.
flip a : $\iota = 2 \rightarrow 3$; flip b : $\iota = 1 \rightarrow 3$; flip c_{in} : $\iota = 1 \rightarrow 5$
All three do, so the carry keeps all three inputs, and the restricted behaviour is matched against the canonical family as MAJORITY.

D. Where the behaviour is already specialised

bound bit 1	0	1	2	3	4	5	6	7
	0	1	1	1	0	1	0	1
bound bit 0	0	1	2	3	4	5	6	7
	0	1	0	1	0	1	0	1

Index $\iota = \ell + 2e + 4x$

bound bit 1: $\mathcal{I} = \{1, 2, 3, 5, 7\}$, essential $\{\ell, e, x\}$
bound bit 0: $\mathcal{I} = \{1, 3, 5, 7\}$, essential $\{\ell\}$
The lower set is the single schema 1^* , anchor 1, offsets $\{0, 2, 4, 6\}$. e and x are free, so no flip of either crosses the set, and the method returns the wire.